

DecodeLabs Internship – Data Science Track

This repository contains my project submissions for the **Data Science Internship at DecodeLabs** (Batch 2026).

Project 1: Advanced EDA & Feature Engineering

Folder: `Project1_EDA_FeatureEngineering/`

Overview

Cleaned and transformed the raw **Titanic dataset** into a mathematically sound, ML-ready dataset by handling missing values, neutralizing outliers, and engineering new predictive features.

What was done

- **Missing Values:** Handled using different strategies depending on missingness percentage:
 - `Age` (20% missing) → filled with median

- `Embarked` (0.2% missing) → rows dropped
- `Cabin` (77% missing) → converted into a binary feature `Has_Cabin`, original column dropped
- **Outliers:** Detected in the `Fare` column using the IQR method and capped (Winsorized) instead of deleted — reduced max fare from 512 to ~65 while preserving all rows.
- **Feature Engineering:** Created 3 new features:
 - `FamilySize` — total family members aboard
 - `IsAlone` — flags solo travelers
 - `Title` — extracted from passenger names (Mr, Mrs, Miss, etc.)

Files in this folder

File	Description
<code>titanic_cleaned.csv</code>	Final cleaned dataset
<code>Project1_notebook.ipynb</code>	Full Google Colab notebook with code
<code>Project1_Summary.md</code>	Detailed write-up of methods and reasoning

Tools Used

Python, Pandas, NumPy, Google Colab